

NPTEL Course on "Computer System Optimization for ML"

Lecture #52

NPTEL

Agenda

- **Performance Characterization**

NPTEL

What is Performance Characterization?

- Gemini says “Performance characterization is the process of **measuring, analyzing, and quantifying** how a **system**, device, or algorithm **behaves under various conditions**. Its primary goal is to establish baselines, identify bottlenecks, and verify whether a product meets its functional and efficiency requirements.”

NPTEL

Why Study Performance Characterization?

- Gemini says “Studying performance characterization is essential because it **bridges the gap between theoretical system design and real-world execution**. It empowers engineers to **identify bottlenecks, optimize resource allocation, and ensure robust, cost-effective operations** in everything from computing algorithms to physical infrastructure.”

NPTEL

Characterization of Deep Learning Workload [1]

- **Include a broader set of benchmarks representing the space**
  - State-of-the-art
- **Wide variety of target tasks**
  - Speech, Language, image/vision
- **Different learning models**
  - Supervised, unsupervised, reinforcement
- **Different types of Neuronal style and connections**
  - Fully-connected, convolutional,
  - Linear, recurrent, residual

[1] Adolf, Robert, et al. "Fathom: Reference workloads for modern deep learning methods." *2016 IEEE International Symposium on Workload Characterization (IISWC)*. IEEE, 2016.

Workloads Considered

Model Name	Year and Ref	Neuronal Style	Layers	Learning Task	Dataset
seq2seq	2014 [43]	Recurrent	7	Supervised	WMT-15 [7]
memnet	2015 [42]	Memory Network	3	Supervised	bAbI [45]
speech	2014 [25]	Recurrent, Full	5	Supervised	TIMIT [22]
autoenc	2014 [32]	Full	3	Unsupervised	MNIST [34]
residual	2015 [27]	Convolutional	34	Supervised	ImageNet [20]
vgg	2014 [41]	Convolutional, Full	19	Supervised	ImageNet [20]
alexnet	2012 [33]	Convolutional, Full	5	Supervised	ImageNet [20]
deepq	2013 [36]	Convolutional, Full	5	Reinforcement	Atari ALE [5]

NDTIL

Workloads Considered

Model Name	Year and Ref	Neuronal Style	Layers	Learning Task	Dataset

- Reference implementation
- Does not capture pre- and post-processing
- Reproducibility
  - Program
  - Input data set
- Implementation in Tensorflow framework
  - Other frameworks (Theano, Caffe, PyTorch) are similar !
  - Does choice of framework matter in performance characterization ?

NPTEL

Fathom Workload

Model Name	Year and Ref	Neuronal Style	Purpose and Legacy
seq2seq	2014 [43]	Recurrent	Direct language-to-language sentence translation. State-of-the-art accuracy with a simple, language-agnostic architecture.
memnet	2015 [42]	Memory Network	Facebook's memory-oriented neural system. One of two novel architectures which explore a topology beyond feed-forward lattices of neurons.
speech	2014 [25]	Recurrent, Full	Baidu's speech recognition engine. Proved purely deep-learned networks can beat hand-tuned systems.
autoenc	2014 [32]	Full	Variational autoencoder. An efficient, generative model for feature learning.
residual	2015 [27]	Convolutional	Image classifier from Microsoft Research Asia. Dramatically increased the practical depth of convolutional networks. ILSVRC 2015 winner.
vgg	2014 [41]	Convolutional, Full	Image classifier demonstrating the power of small convolutional filters. ILSVRC 2014 winner.
alexnet	2012 [33]	Convolutional, Full	Image classifier. Watershed for deep learning by beating hand-tuned image systems at ILSVRC 2012.
deepq	2013 [36]	Convolutional, Full	Atari-playing neural network from DeepMind. Achieves superhuman performance on majority of Atari2600 games, without any preconceptions.

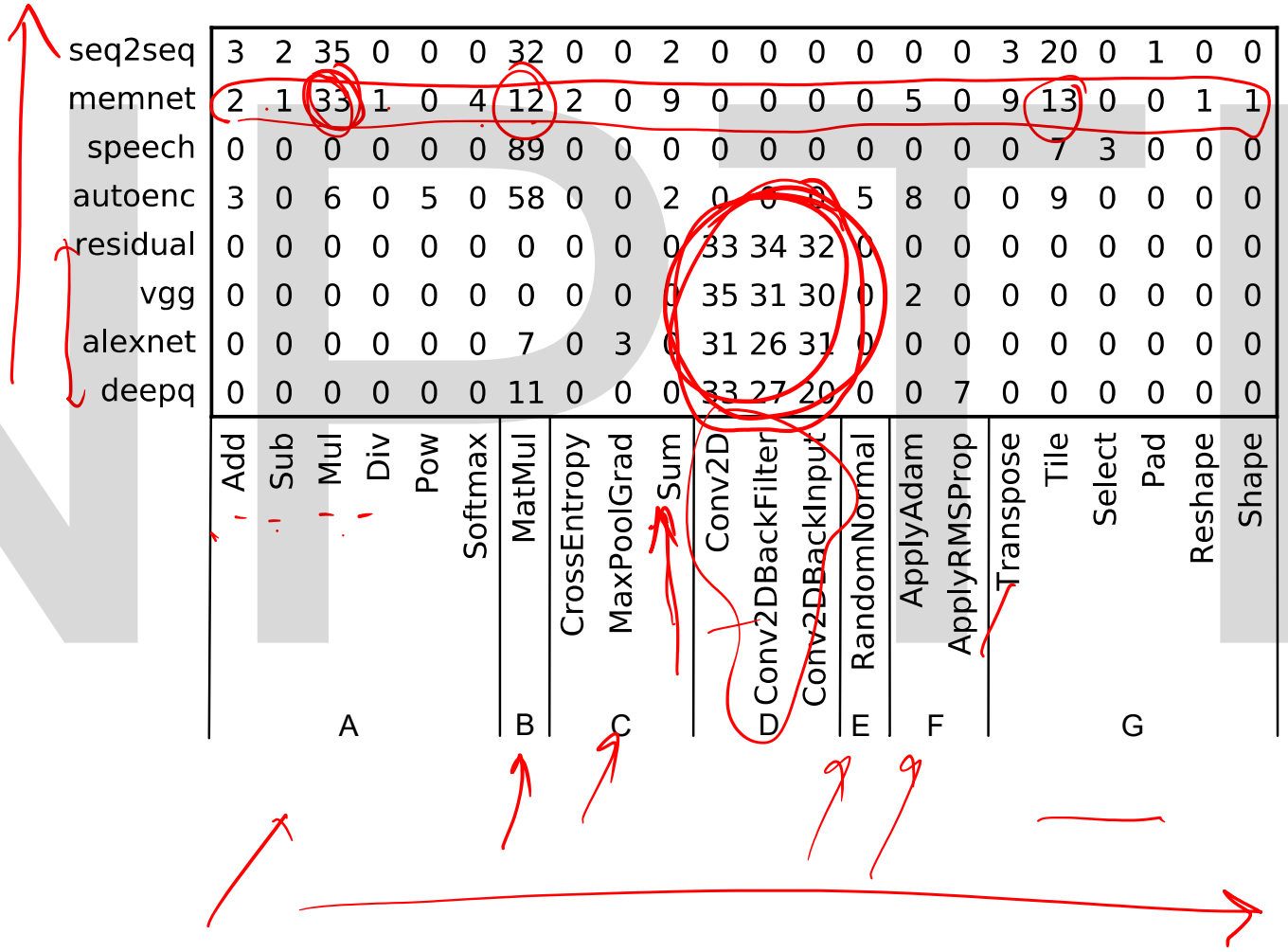
NPTEL

Performance Characterization

- Low-level profiling (using perf or PMC) is possible
  - how to relate them to different layers
- Uses Tensorflow measurement tools
- Operation-level profiling

NPTEL

Operation Type Profiling



	Add	Sub	Mul	Div	Pow	Softmax	MatMul	CrossEntropy	MaxPoolGrad	Sum	Conv2D	Conv2DBackFilter	Conv2DBackInput	RandomNormal	ApplyAdam	ApplyRMSProp	Transpose	Tile	Select	Pad	Reshape	Shape
seq2seq	3	2	35	0	0	0	32	0	0	2	0	0	0	0	0	0	3	20	0	1	0	0
memnet	2	1	33	1	0	4	12	2	0	9	0	0	0	0	5	0	9	13	0	0	1	1
speech	0	0	0	0	0	0	89	0	0	0	0	0	0	0	0	0	0	7	3	0	0	0
autoenc	3	0	6	0	5	0	58	0	0	2	0	0	0	5	8	0	0	9	0	0	0	0
residual	0	0	0	0	0	0	0	0	0	0	33	34	32	0	0	0	0	0	0	0	0	0
vgg	0	0	0	0	0	0	0	0	0	0	35	31	30	0	2	0	0	0	0	0	0	0
alexnet	0	0	0	0	0	0	7	0	3	0	31	26	31	0	0	0	0	0	0	0	0	0
deepq	0	0	0	0	0	0	11	0	0	0	33	27	20	0	0	7	0	0	0	0	0	0
	A						B	C			D			E	F		G					

Group	Op Class
A	Elementwise Arithmetic
B	Matrix Operations
C	Reduction and Expansion
D	Convolution
E	Random Sampling
F	Optimization
G	Data Movement

Operation Type Profiling

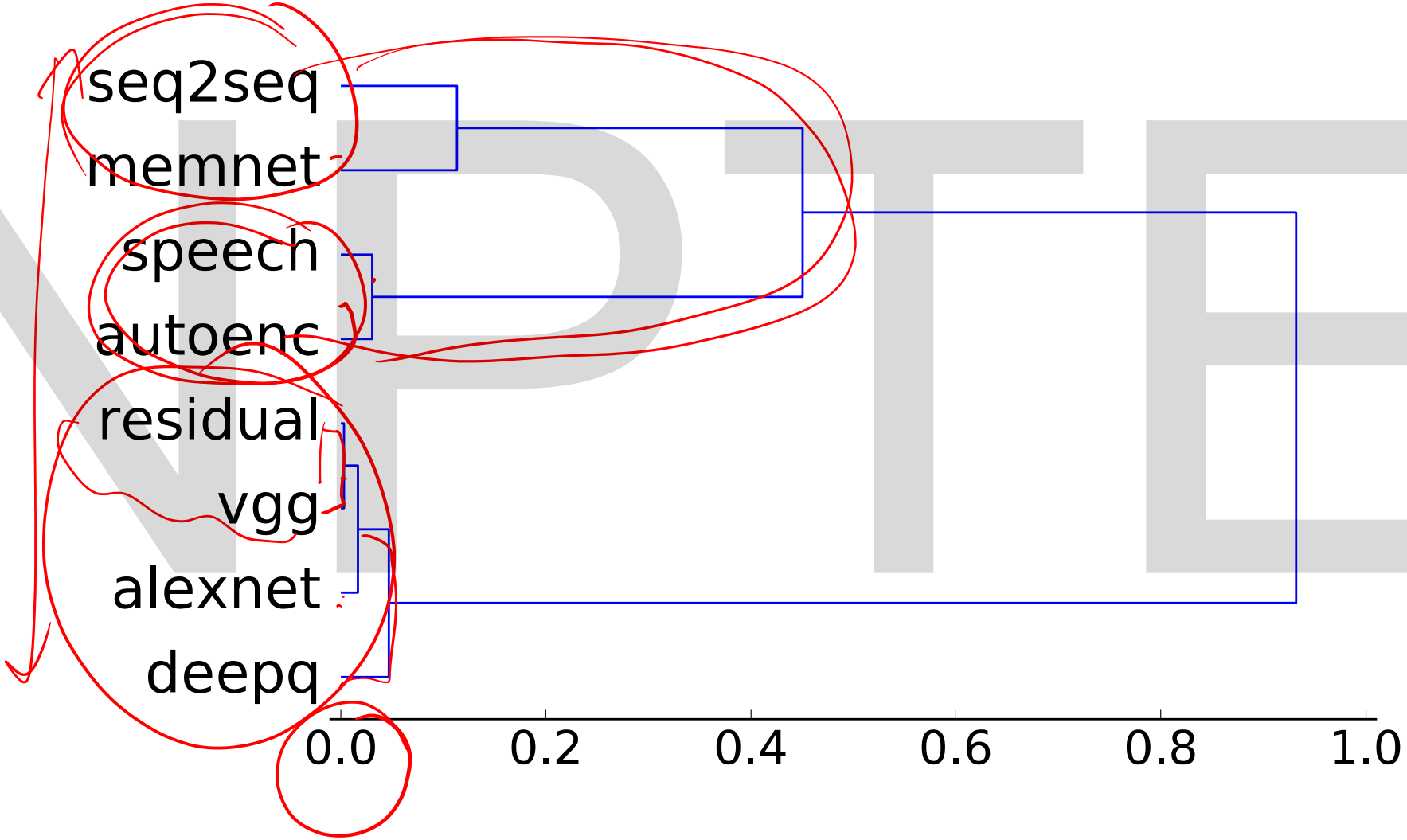
seq2seq	3	2	35	0	0	0	32	0	0	2	0	0	0	0	0	0	3	20	0	1	0	0
memnet	2	1	33	1	0	4	12	2	0	9	0	0	0	0	5	0	9	13	0	0	1	1
speech	0	0	0	0	0	0	89	0	0	0	0	0	0	0	0	0	0	7	3	0	0	0

Group	Op Class
A	Elementwise Arithmetic
G	Data Movement

- A few operations dominate the execution of each model
  - Convolution in CNN, MatMul in Fully Connected Network
- **Percentage of time spent for different types of operations can be used as characteristics of the application**
  - To understand similarity of application

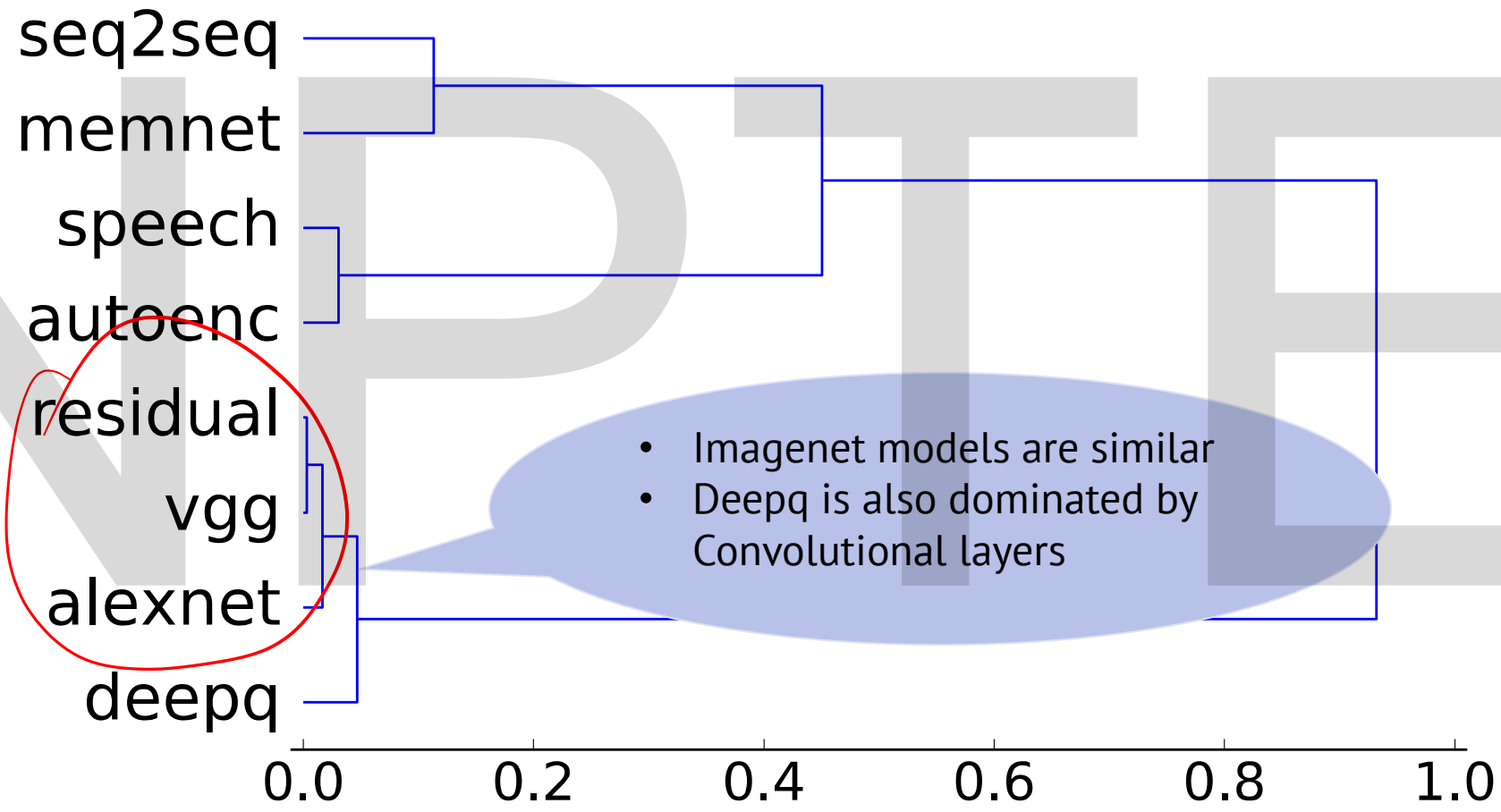
NDIL

Performance Similarity of Benchmarks



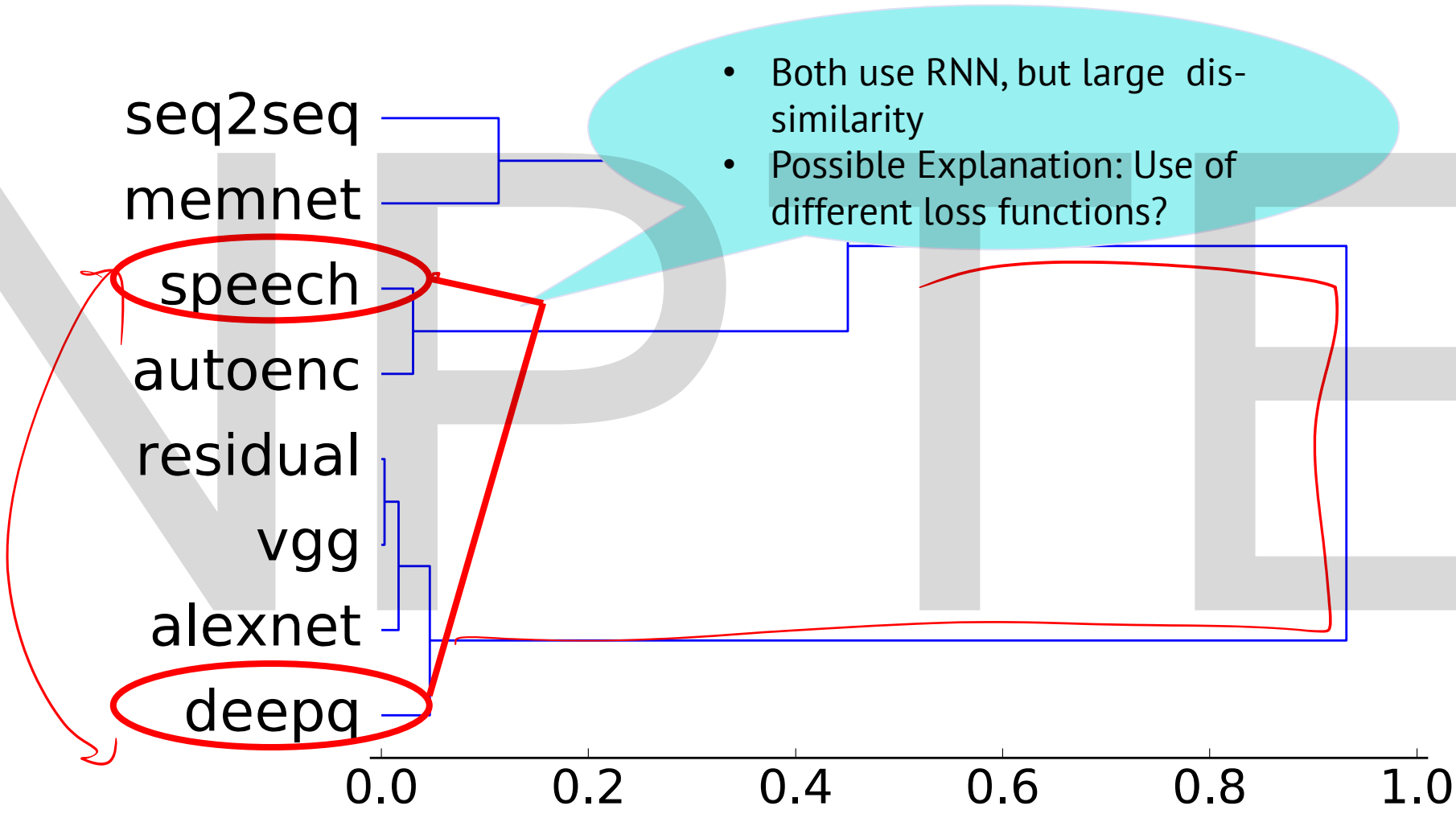
NPTEL

Performance Similarity of Benchmarks



NPTEL

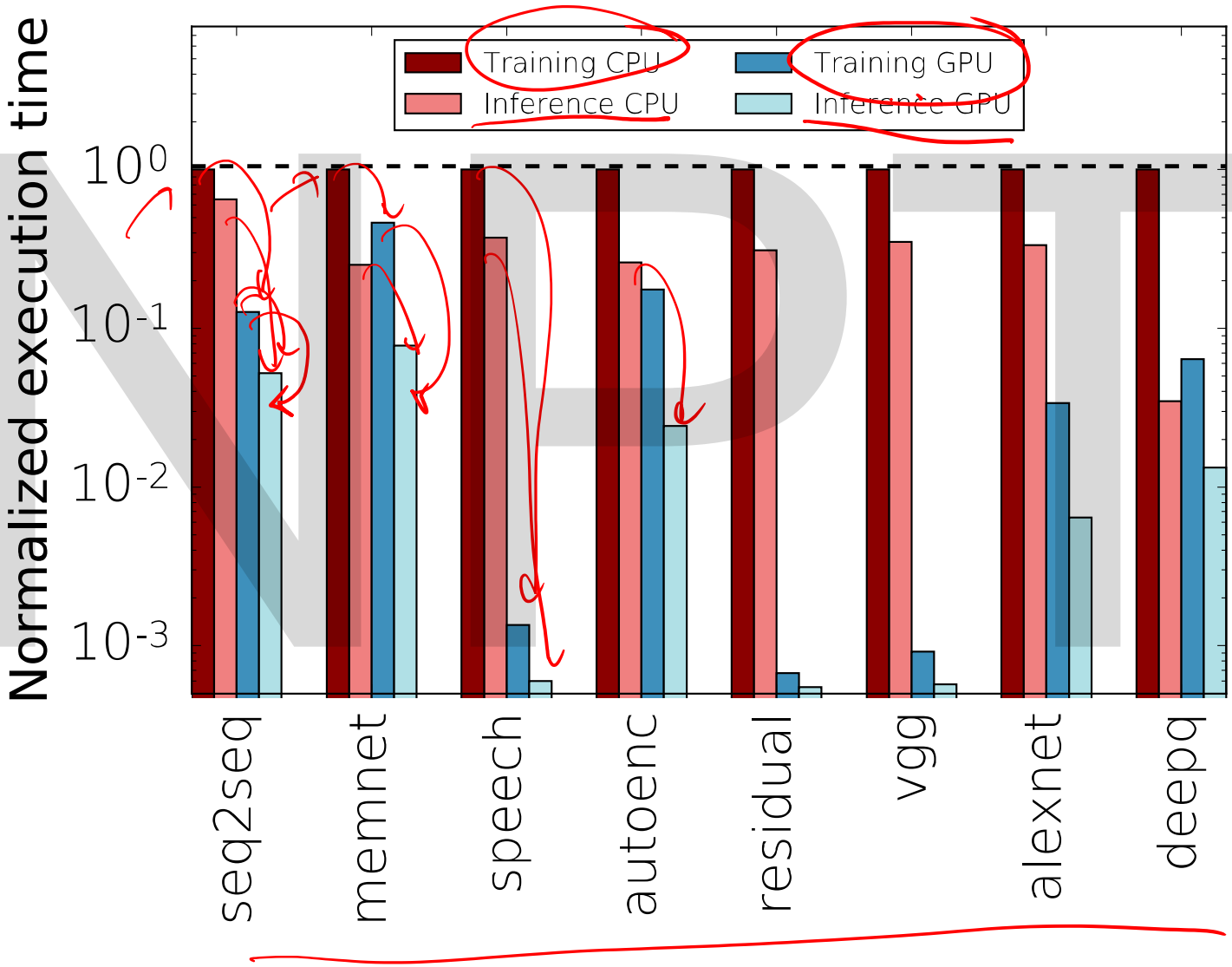
Performance Similarity of Benchmarks



- Both use RNN, but large dis-similarity
- Possible Explanation: Use of different loss functions?

NPTEL

Training vs. Inference Execution Times

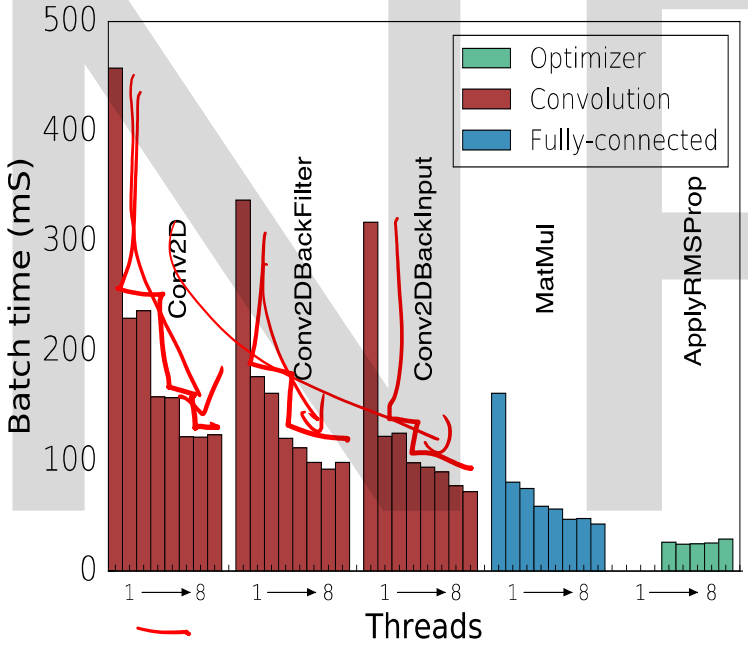


- Training
  - Forward and backward pass
  - Multiple batches and epochs

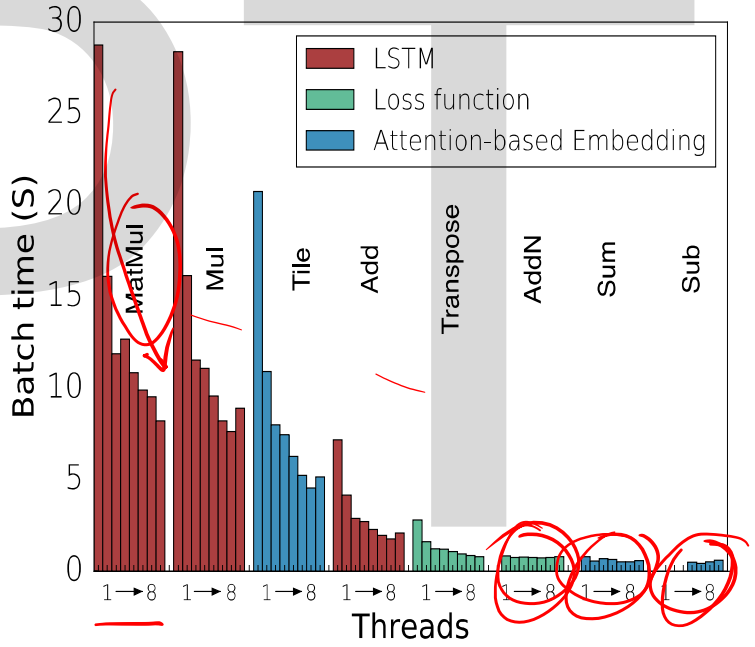
INTEL

Parallelism (intra-operation)

- **Parallelism exploited by assigning more threads to parallel libraries**
  - Convolution, MatMul, …



(a) Operation type scaling in `deepq`

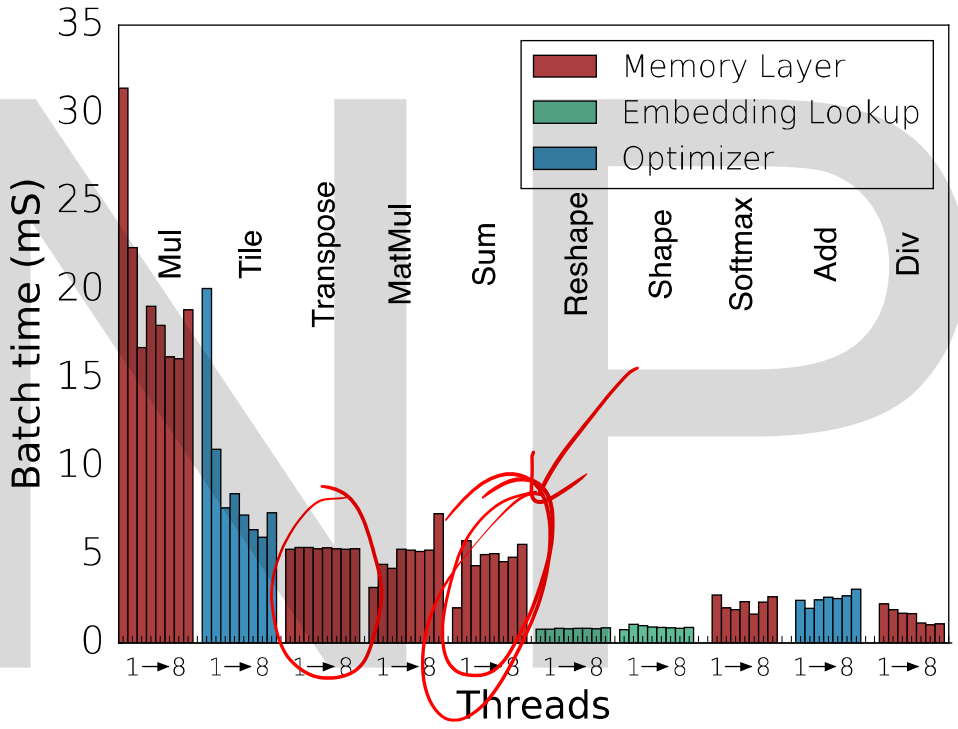


(b) Operation type scaling in `seq2seq`

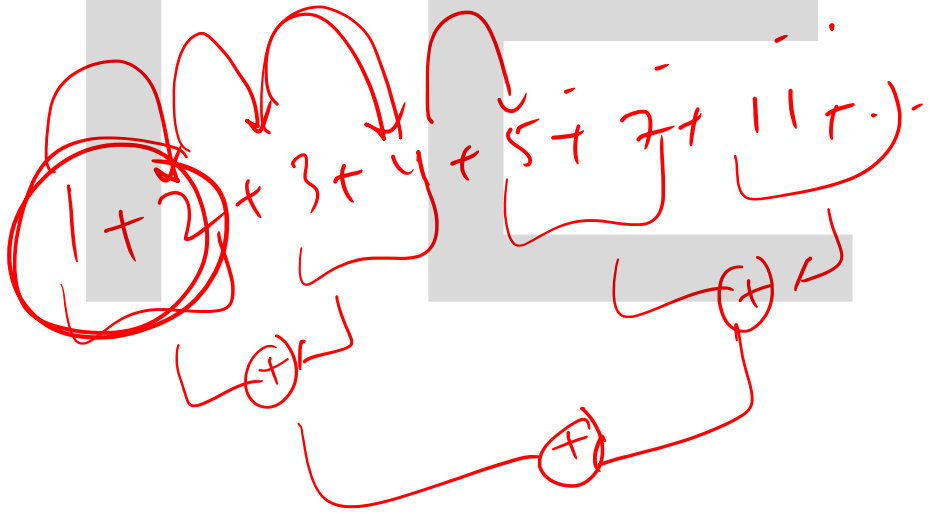
Observations

- Convolution: Reduction with increasing threads. Staircase patterns
- Matmul: Similar but less pronounced staircase
- Element-wise operation behave differently

Parallelism (intra-operation)



- **Observations**
  - Small/no reduction in transpose
  - Execution time increases with number of threads for 'sum'





THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)